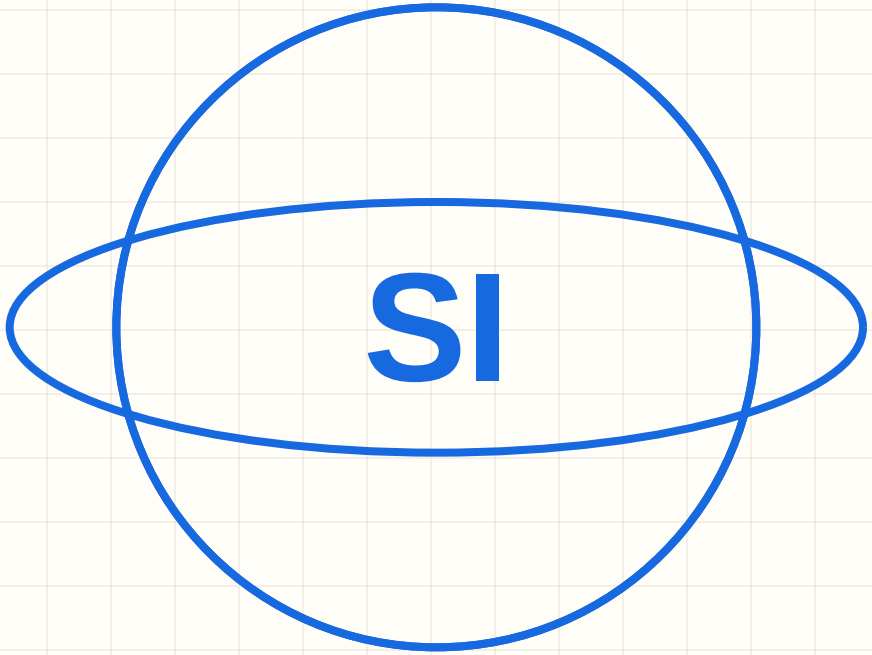
CLASS 11 PHYSICS / DEMO LECTURE

# Every measurement tells a story.

Physical quantities, SI units and the art of converting between them.

CBSE FOUNDATION  NEET READY  INTERACTIVE DEMO




Teacher: _____ Date: _____

OPENING STORY / REAL CONSEQUENCE

# A $125 million unit mistake

The Mars Climate Orbiter was lost because two engineering teams used different unit systems.

## Team A

**pound-force seconds**

Imperial measurement output

!=

## Team B

**newton seconds**

SI measurement expected

**Discuss:** Where could a unit mistake be dangerous in everyday life?

QUICK CLASS CHALLENGE

# Can it be measured objectively?

Circle the physical quantities. Cross out the others.

| | | |
|---|---|---|
| **Length** | **Happiness** | **Temperature** |
| **Beauty** | **Mass** | **Time** |

**Physical quantity = numerical value x unit**

# Base quantities build the language. Derived quantities build the physics.

INDEPENDENT

## Base quantities

Length

Mass

Time

Electric current

Temperature

CONSTRUCTED

## Derived quantities

Area = length x length

Speed = distance / time

Force = mass x acceleration

**Your turn:** Is velocity a base quantity or a derived quantity? Explain.

# Seven agreed foundations. One scientific language

| | | |
|---|---|---|
| m | **Length** | metre |
| kg | **Mass** | kilogram |
| s | **Time** | second |
| A | **Current** | ampere |
| K | **Temperature** | kelvin |
| mol | **Amount** | mole |
| cd | **Intensity** | candela |

Example: _____

Board note: Kelvin uses K, not degree K. Unit symbols are never plural.

BUILD, DO NOT MEMORISE

# A derived unit comes directly from its equation.

**Force = mass x acceleration**

| MASS | | ACCELERATION | | FORCE | | NAME |
|---|---|---|---|---|---|---|
| kg | -> | m s^-2 | -> | kg m s^-2 | -> | newton (N) |

**Board task:** Derive the SI unit of speed and acceleration.

# Multiply by conversion factors equal to one.

Write the units on every line. Cancel them like algebraic factors.

$$72\ \text{km h}^{-1} \times \frac{1000\ \text{m}}{1\ \text{km}} \times \frac{1\ \text{h}}{3600\ \text{s}} = 20\ \text{m s}^{-1}$$

**SPEED SHORTCUT**

**km/h -> m/s: multiply by 5/18 | m/s -> km/h: multiply by 18/5**

Use the whiteboard pen to cancel km and h above.

STUDENT CHALLENGE / 45 SECONDS

# A car moves at 90 km h^-1. What is its speed in m s^-1?

A 15 m s^-1

B 20 m s^-1

C 25 m s^-1

D 32.4 m s^-1

**Show your work:**

FOUR MISTAKES TO ELIMINATE

# Correct method earns marks and

| | | |
|---|---|---|
| 1 | Number without a unit | Write 20 m s^-1 |
| 2 | Writing kgs or secs | Use kg and s |
| 3 | Wrong conversion direction | km/h to m/s: x 5/18 |
| 4 | Too many calculator digits | Respect given precision |

Estimate first: the converted answer should make physical sense.

PRACTICE BOARD / WRITE DIRECTLY HERE

# Three questions. Show every unit.

**1. Convert 36 km h^-1 to m s^-1.**

**2. Write the SI unit of force using base units.**

**3. A measurement is written as 12.5. What essential information is missing?**

WRITE / DISCUSS / CORRECT

ONE-MINUTE UNDERSTANDING CHECK

# Finish these three sentences.

1. **A physical quantity must contain ...**

2. **The SI base unit of mass is ...**

3. **To convert km/h to m/s, I ...**

**Do not reveal the answers until students commit to a response.**

ORBIT LEARNING SYSTEM

# The lecture ends. The learning loop does not.

1
**Learn**
60-minute lesson

->

2
**Attempt**
5-10 questions

->

3
**Analyse**
Topic-wise data

->

4
**Correct**
Next-day review

->

5
**Continue**
Confident progress

**Next: complete the daily test in the Orbit Physics app**

**Tomorrow's lesson begins with evidence, not assumptions.**